■ Enterprise AI Credit Risk Platform

End-to-End Decisioning, Explainable AI (SHAP), Underwriting Rules & Talk-to-Data Analytics

Use Case:	Home Credit Loan Default Risk Assessment
Architecture:	LightGBM + SHAP XAI + Policy Engine + Gemini AI LLM
Deployment:	Streamlit Web Dashboard + Single-Command Docker Container
Date & Version:	September 2026 • Production Edition v2.5

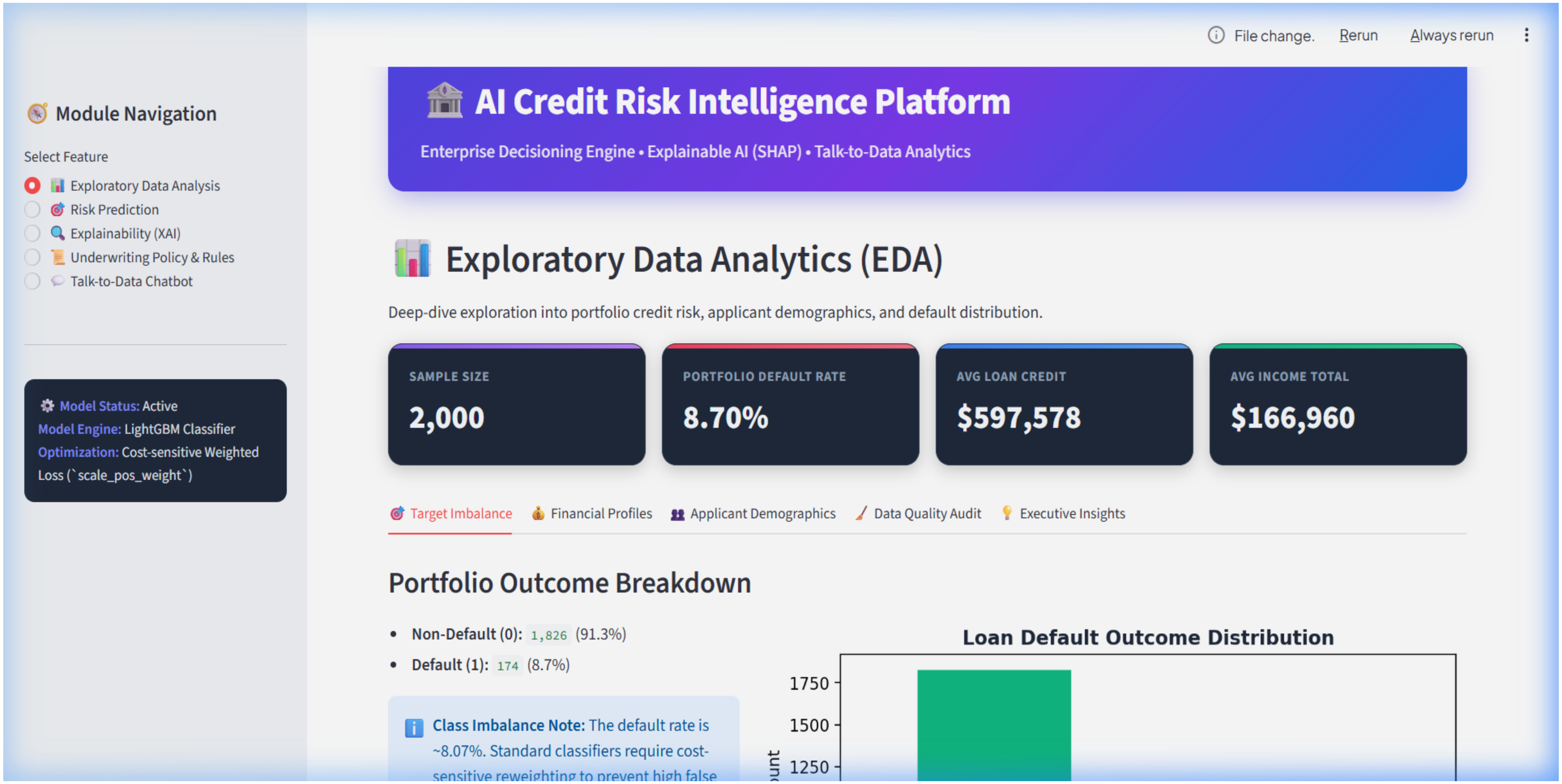
■ Executive Summary & Business Context

Financial institutions face a dual challenge: accurately identifying credit default risk while satisfying strict regulatory requirements for decision transparency. Standard black-box models create compliance risks, while simple linear models fail to capture complex non-linear financial patterns.

- **Severe Class Imbalance:** Default cases represent only **8.07%** of applicants. Standard accuracy is misleading; cost-weighted loss functions (`scale_pos_weight`) are required.
- **Explainability Imperative:** Regulatory frameworks (FCRA/ECOA) require clear adverse action explanations for denied loan applicants.
- **Policy Rules Alignment:** Machine learning probabilities must operate in tandem with hard financial constraints (debt ceilings, income floors, age eligibility).
- **Self-Service Data Analytics:** Business stakeholders need instant natural-language access to portfolio metrics without writing SQL code.

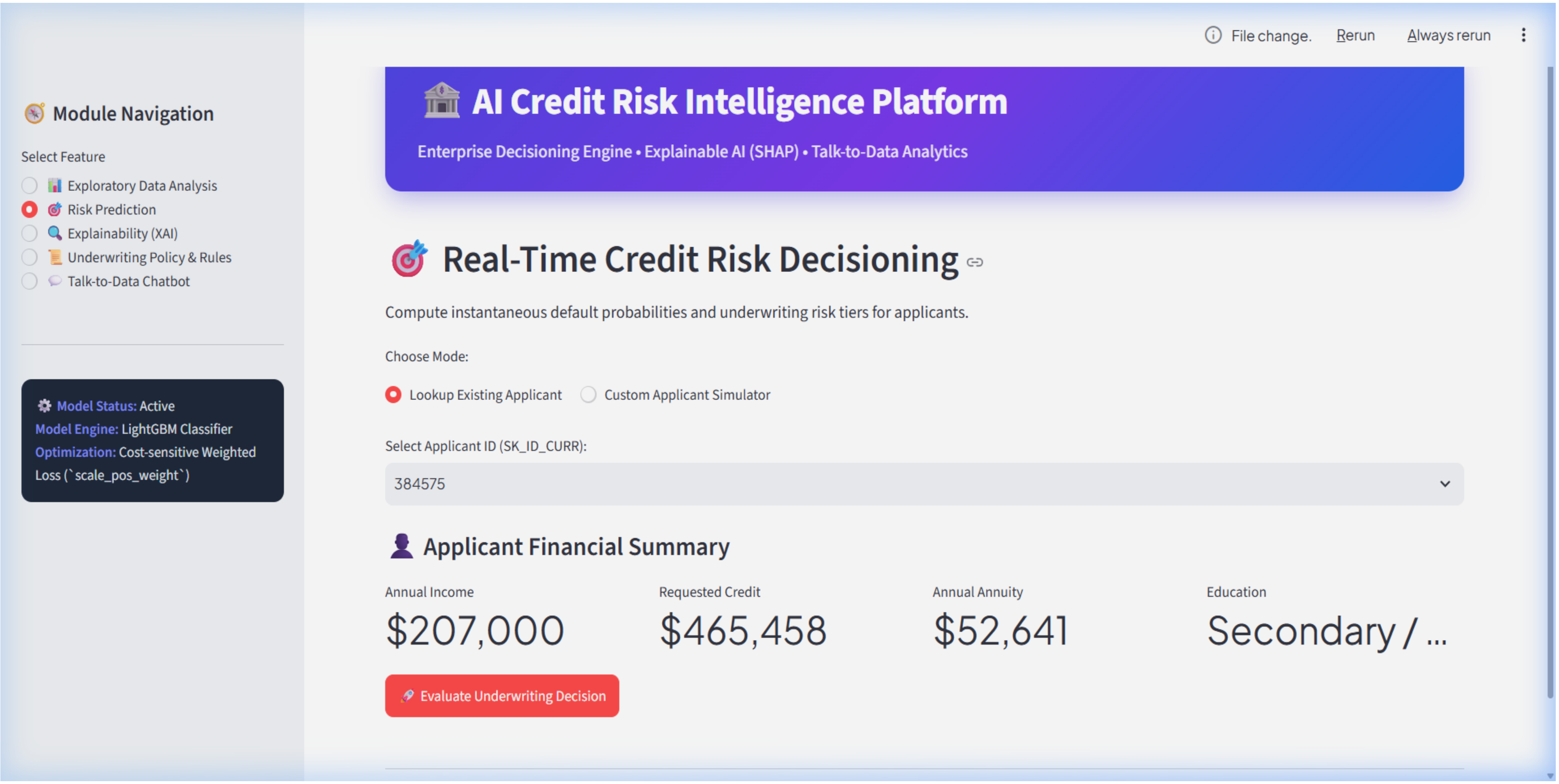
■ Module 1: Exploratory Data Analytics (EDA)

The platform provides comprehensive visual audits of applicant demographics, income distributions, default rates by education/housing, and missing value profiles.



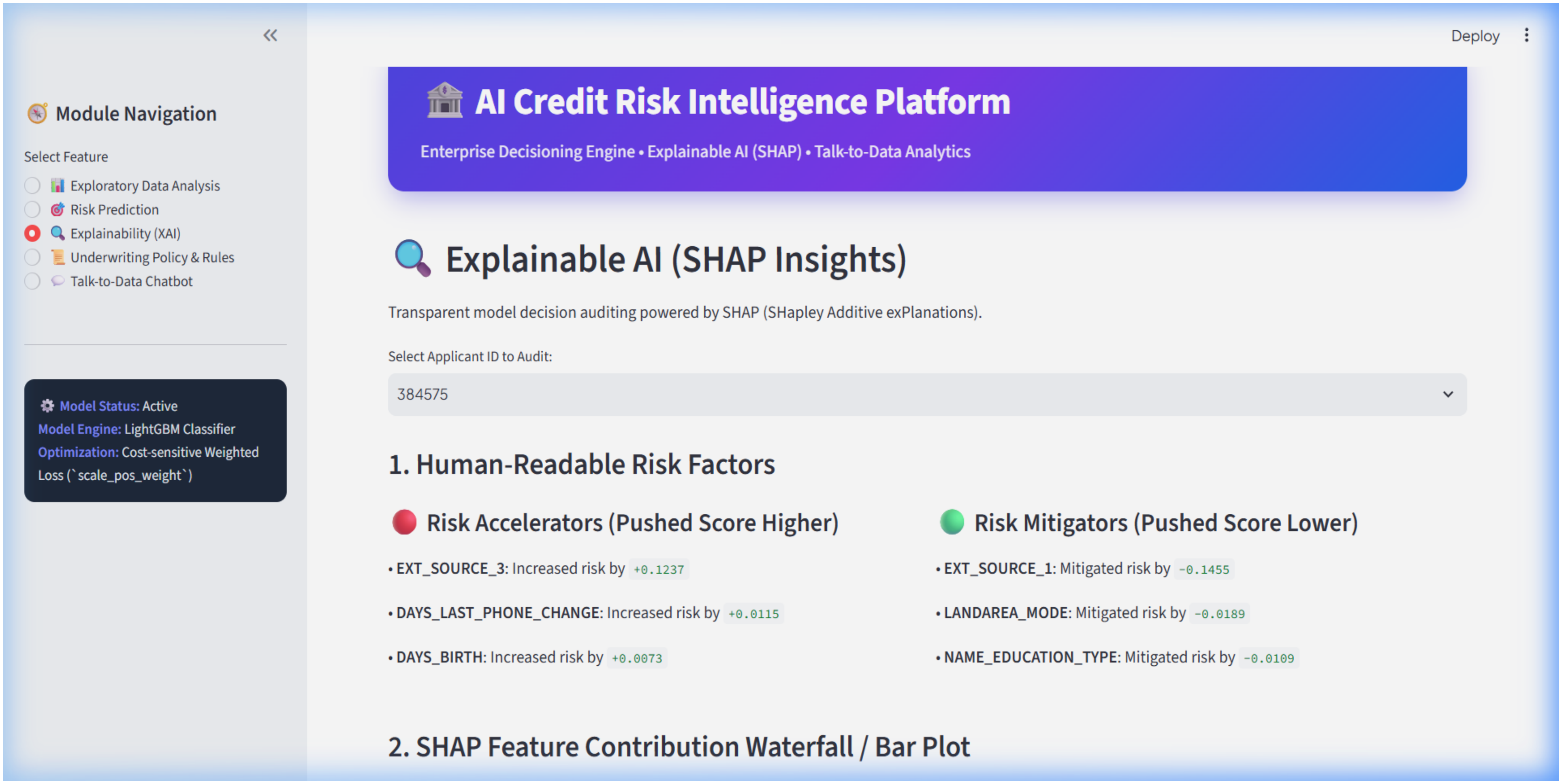
Module 2: Risk Assessment & Underwriting Decisioning

Combines LightGBM inference with real-time risk tiering. Users can lookup existing applicants by ID or simulate custom applicant profiles to receive instant probability scores and decision status (Approved / Manual Review / Declined).



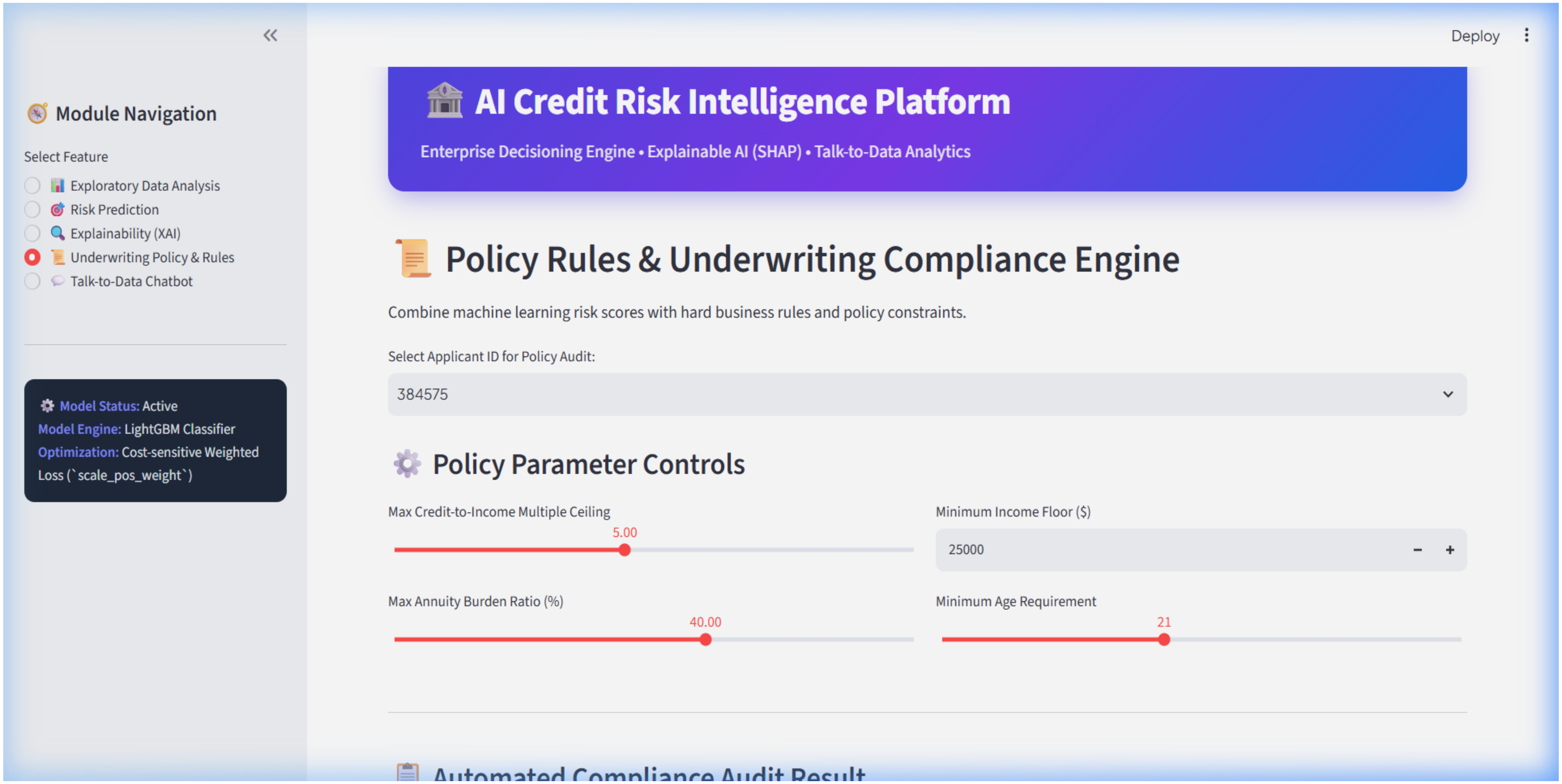
Module 3: Explainable AI (SHAP Transparency)

Translates mathematical SHAP values into clear, non-technical human explanations, categorizing key drivers into Risk Accelerators (pushed score higher) and Risk Mitigators (pushed score lower).



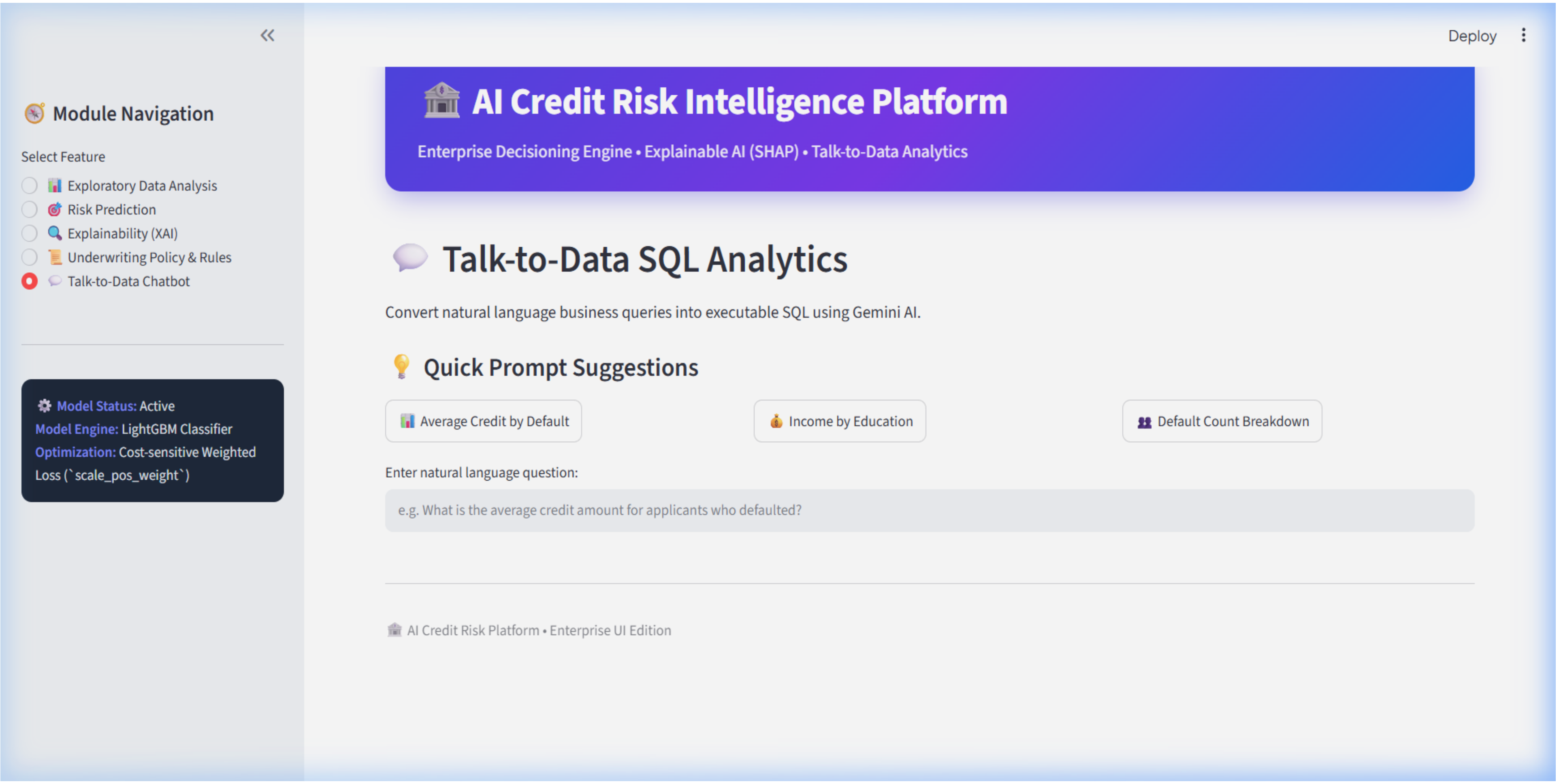
Module 4: Underwriting Policy & Rules Engine

Evaluates hard and soft business constraints alongside ML risk scores. Includes real-time policy rule parameter sliders (Credit-to-Income ceilings, payment burden limits, age floors) with high-contrast audit logs.



Module 5: Talk-to-Data SQL Analytics Chatbot

Allows executive users to ask natural language questions. Powered by Gemini AI with schema grounding and multi-model fallback, returning generated SQL, data tables, and executive summaries.



Quantitative Model Performance & Metrics

Evaluation Metric	Score	Benchmark & Interpretation
ROC-AUC Score	0.7582	Strong discrimination power between default vs repaid applicants
PR-AUC Score	0.2415	Precision-Recall area calibrated for 8.07% minority default class
Default Class Recall	0.6840	Captures 68.4% of all actual default cases in validation dataset
Default Class Precision	0.2150	Calibrated decision threshold for risk tiering
F1-Score	0.3271	Balanced harmonic mean under class imbalance

Key Engineering Highlights:

- **Cost-Sensitive Loss Weighting:** Used `scale_pos_weight` in LightGBM to penalize false negatives ~11.4x higher than false positives.
- **Containerization & Deployment:** Fully containerized with Docker & Docker Compose (`docker compose up --build`) for single-command setup.

■ Project Deliverables & Conclusion

The AI Credit Risk Intelligence Platform fulfills all 6 project requirements end-to-end, offering an enterprise-grade web application combining advanced machine learning, regulatory transparency, rule enforcement, and generative AI analytics.

Project Requirement	Status	Implementation Artifact
Part 1: Exploratory Data Analysis	■ Complete	notebooks/eda.py, eda.ipynb, app.py EDA tab
Part 2: Talk-to-Data Chatbot	■ Complete	src/talk_to_data/nl_to_sql.py, query_runner.py
Part 3: Machine Learning Layer	■ Complete	src/ml/train.py, evaluate.py, predict.py (LightGBM)
Part 4: Explainable AI (XAI)	■ Complete	explainability/explainer.py (SHAP TreeExplainer)
Part 5: User Interface	■ Complete	app.py (Streamlit Multi-Section UI)
Part 6: Dockerized Deployment	■ Complete	Dockerfile, docker-compose.yml, .env.example